

MAE 140 Winter 12 Homework #3

T R & T, 3.1, 3.3, 3.6, 3.7, 3.9, 3.10, 3.12, 3.15 b) and d), 3.71, 3.74

Due: 01/31/2012

3-1 (a) Formulate node-voltage equations for the circuit in Figure P3-1.

- (b) Solve these equations for v_A and v_B .
 (c) Use these results to find v_x and i_x .

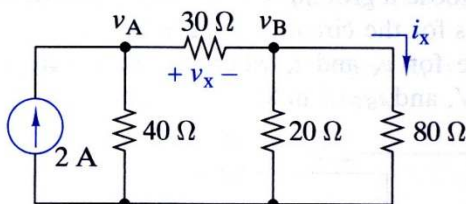


FIGURE P3-1

3-3 (a) Formulate node-voltage equations for the circuit in Figure P3-3.

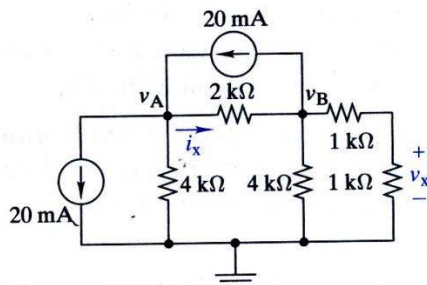


FIGURE P3-3

- (b) Solve these equations for v_A and v_B .
 (c) Use these results to find v_x and i_x .

3-6 (a) Formulate node-voltage equations for the circuit in Figure P3-6.

- (b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = 10\text{ k}\Omega$ and $v_s = 24\text{ V}$.

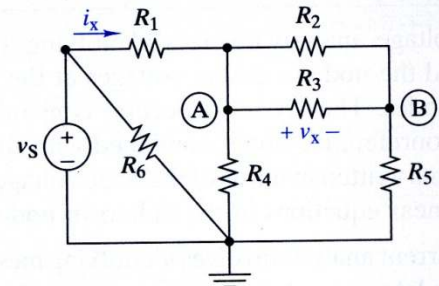


FIGURE P3-6

3-7 (a) Formulate node-voltage equations for the circuit in Figure P3-7.

- (b) Use MATLAB to find symbolic expressions for the node voltages in terms of the parameters in the circuit.
 (c) Find numeric values for v_A , v_B , and v_C when $R_1 = 1.5\text{ k}\Omega$, $R_2 = 2.2\text{ k}\Omega$, $R_3 = 3.3\text{ k}\Omega$, $R_4 = 4.7\text{ k}\Omega$, and $i_{S1} = i_{S2} = 2\text{ mA}$.
 (d) Use OrCAD to verify that your solution to part (c) is correct.

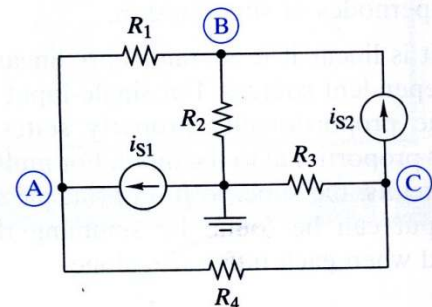


FIGURE P3-7

3-9 (a) Formulate mesh-current equations for the circuit in Figure P3-9.

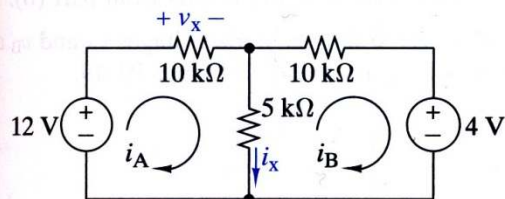


FIGURE P3-9

- (b) Solve for i_A and i_B .
 (c) Use these results to find v_x and i_x .

3-10(a) Formulate mesh-current equations for the circuit in Figure P3-10.

- (b) Solve for i_A and i_B .
 (c) Use these results to find v_x and i_x .

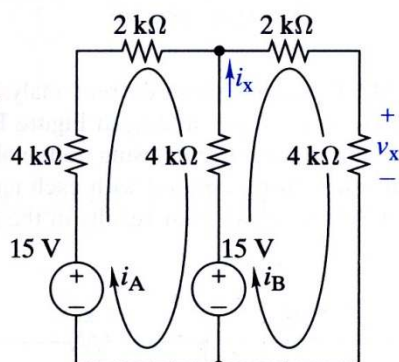


FIGURE P3-10

3-12(a) Formulate mesh-current equations for the circuit in Figure P3-12.

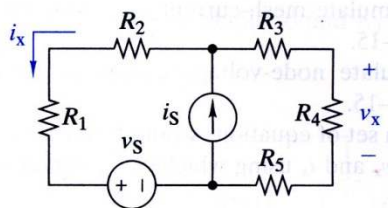


FIGURE P3-12

- (b) Solve for v_x and i_x when $R_1 = 200\ \Omega$, $R_2 = 300\ \Omega$, $R_3 = 50\ \Omega$, $R_4 = 250\ \Omega$, $R_5 = 200\ \Omega$, $i_s = 100\ \text{mA}$, and $v_s = 25\ \text{V}$.
 (c) Find the total power dissipated in the circuit.

3-15(a) Formulate mesh-current equations for the circuit in Figure P3-15.

(b) Formulate node-voltage equations for the circuit in Figure P3-15.

- (c) Which set of equations would be easier to solve? Why?
 (d) Find v_x and i_x using whichever method you prefer.

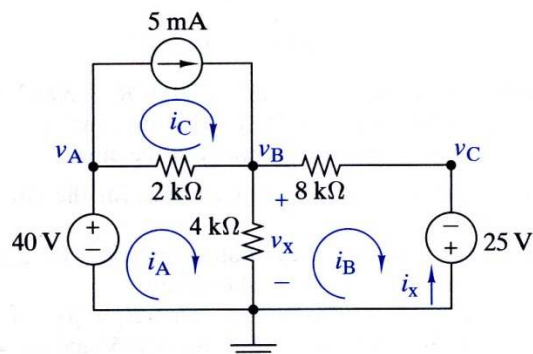
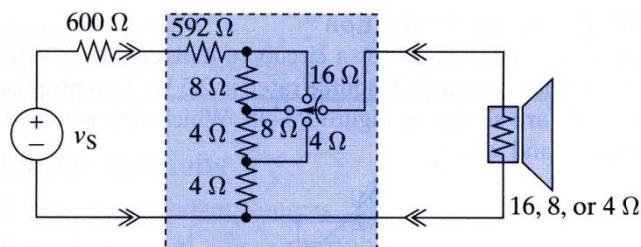


FIGURE P3-15

3-71 **Audio Speaker Resistance-Matching Network**

A company is producing an interface network that they claim will result in R_{IN} of $600\ \Omega \pm 2\%$ and R_{OUT} of 16, 8, or $4\ \Omega \pm 2\%$ —depending on whether the connected speakers are 16, 8, or $4\ \Omega$ —selectable via a built-in switch. Prove or disprove their claim.



Audio matching network

FIGURE P3-71

3-74 **Battery Design**

A satellite requires a battery with an open-circuit voltage $v_{OC} = 36\ \text{V}$ and a Thévenin resistance $R_T \leq 10\ \Omega$. The battery is to be constructed using series and parallel combinations of one of two types of cells. The first type has $v_{OC} = 9\ \text{V}$, $R_T = 4\ \Omega$, and a weight of 40 grams. The second type has $v_{OC} = 4\ \text{V}$, $R_T = 0.5\ \Omega$, and a weight of 15 grams. Design a minimum-weight battery that meets the open-circuit voltage and Thévenin resistance requirements.